Sobolev Inequalities

<u>Q:</u> Are there exponents $q \in [1, \infty)$ st for all $u \in W^{1,p}(\mathbb{R}^n)$, $u \in L^q(\mathbb{R}^n)$ and

$$\|u\|_{L^q(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)}$$

for some constant $C > 0$?

Suppose there is such a q. Then

$$\left(\int_{\mathbb{R}^n} |u|^q \, dx\right)^{1/q} \leq C \left(\int_{\mathbb{R}^n} |Du|^p \, dx\right)^{1/p}.$$

Consider the scaling

$$u_\lambda(x) = u(\lambda x), \quad \lambda > 0.$$

We have

$$\int_{\mathbb{R}^n} |Du_\lambda(x)|^p \, dx = \int_{\mathbb{R}^n} \lambda^p |Du(\lambda x)| \, dx = \int_{\mathbb{R}^n} \lambda^{p-n} |Du(y)|^p \, dy$$

$$y = \lambda x$$

$$dy = \lambda^n dx$$

$$= \lambda^{p-n} \int_{\mathbb{R}^n} |Du|^p \, dx$$

$$u_\lambda(x) = u(\lambda x)$$

$$Du_\lambda(x) = \lambda Du(\lambda x)$$

$$\int_{\mathbb{R}^n} |u_\lambda|^q \, dx = \int_{\mathbb{R}^n} |u(\lambda x)|^q \, dx = \lambda^{-n} \int_{\mathbb{R}^n} |u|^q \, dx$$

So

$$\left(\int_{\mathbb{R}^n} |u|^q \, dx\right)^{1/q} \leq C \left(\int_{\mathbb{R}^n} |Du|^p \, dx\right)^{1/p}$$

gives

$$\left(\lambda^{-n} \int_{\mathbb{R}^n} |u|^q \, dx\right)^{1/q} \leq C \left(\lambda^{p-n} \int_{\mathbb{R}^n} |Du|^p \, dx\right)^{1/p}$$

$$\left(\int_{\mathbb{R}^n} |u|^q \, dx\right)^{1/q} \leq C \lambda^{\overset{\alpha}{\frac{n}{q}+1-\frac{n}{p}}} \left(\int_{\mathbb{R}^n} |Du|^p \, dx\right)^{1/p} \qquad \forall \lambda > 0$$

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$A, B > 0$

$A \leq \lambda^{\alpha} B \quad \forall \lambda > 0 \Rightarrow \alpha$

If $\alpha > 0$, letting $\lambda \to 0$ gives $A \leq 0$, so $\alpha \leq 0$. If $\alpha < 0$, letting $\lambda \to \infty$ gives $A \leq 0$, so $\alpha = 0$.

So $\frac{n}{q} + 1 - \frac{n}{p} = 0$ and hence

$$q = \frac{np}{n-p}.$$

For $1 \leq p < n$,

q is called the <u>critical Sobolev exponent</u> and is denoted by p^*:

$$\boxed{p^* = \frac{np}{n-p}} \quad \left(\geq \frac{np}{n} = p \geq 1 \right)$$

Thm (Gagliardo-Nirenberg-Sobolev inequality):

If $1 \leq p < n$, then $\exists$ a constant $C > 0$ st

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)} \quad \forall u \in C_c^1(\mathbb{R}^n).$$

<u>Pf</u>: First we consider the case $p=1$. For simplicity, we take $n=2$. So we want to prove

$$\left(\int_{\mathbb{R}^2} |u|^2 dx\right)^{1/2} \leq C \int_{\mathbb{R}^2} |Du| dx.$$

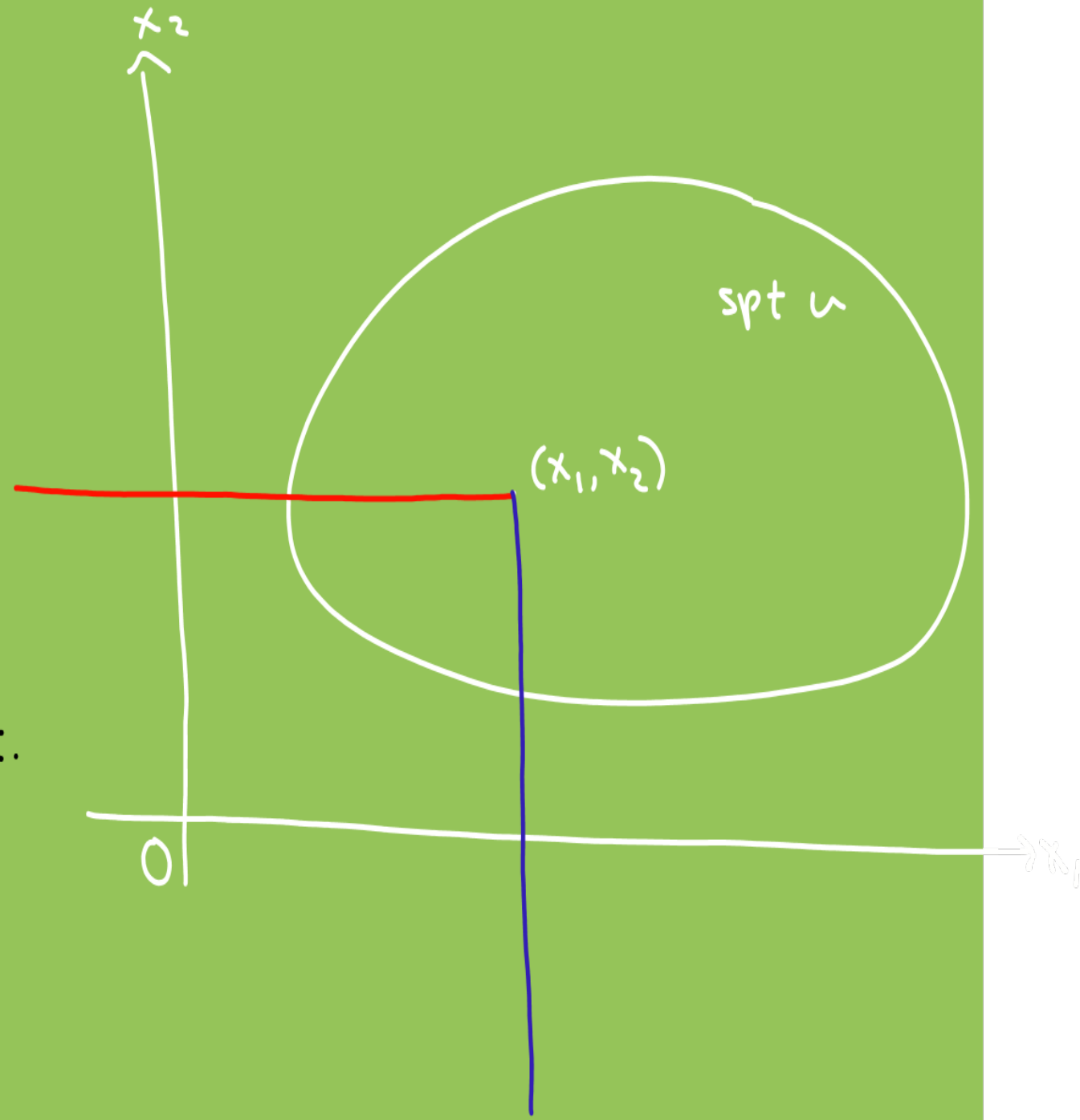
$$\boxed{p^* = \frac{np}{n-p} = 2 \quad \text{when } p=1 \text{ and } n=2}$$

So $W^{1,p}(\mathbb{R}^n) \cap C_c^1(\mathbb{R}^n) \subset L^{p^*}(\mathbb{R}^n)$.

$$u(x_1,x_2) = \int_{-\infty}^{x_1} u_{x_1}(y_1,x_2)\,dy_1$$

$$u(x_1,x_2) = \int_{-\infty}^{x_2} u_{x_2}(x_1,y_2)\,dy_2$$

since u has compact support.



$$|u(x_1,x_2)| = \left| \int_{-\infty}^{x_1} u_{x_1}(y_1,x_2)\,dy_1 \right|$$

$$\leq \int_{-\infty}^{x_1} |u_{x_1}(y_1,x_2)|\,dy_1$$

$$\leq \int_{-\infty}^{\infty} |Du(y_1,x_2)|\,dy_1$$

$$Du = (u_{x_1}, u_{x_2})$$

$$|Du| = \sqrt{u_{x_1}^2 + u_{x_2}^2}$$

$$\geq |u_{x_1}|$$

Similarly,

$$|u(x_1,x_2)| \leq \int_{-\infty}^{\infty} |Du(x_1,y_2)|\,dy_2.$$

Multiplying gives

$$|u|^2 \leq \left(\int_{-\infty}^{\infty} |Du(y_1, x_2)|\, dy_1\right)\left(\int_{-\infty}^{\infty} |Du(x_1, y_2)|\, dy_2\right).$$

So

$$\int_{\mathbb{R}^2} |u|^2\, dx \leq \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} |Du(y_1, x_2)|\, dy_1\right)\left(\int_{-\infty}^{\infty} |Du(x_1, y_2)|\, dy_2\right) dx_1\, dx_2$$

x_2 x_1

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du(y_1, x_2)| \, dy_1 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du(x_1, y_2)| \, dy_2 \, dx_1 \, dx_2$$

$$\int_{\mathbb{R}^2} |Du| \, dx$$

$$= \left(\int_{\mathbb{R}^2} |Du| \, dx \right) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du(y_1, x_2) \, dy_1 \, dx_2 = \left(\int_{\mathbb{R}^2} |Du| \, dx \right)^2$$

Now suppose $1<p<n$. We apply the inequality for $p=1$ to $v := |u|^\gamma$, where $\gamma>1$ is to be determined. Since $u \in C_c^1(\mathbb{R}^n)$ and $\gamma>1$, $v \in C_c^1(\mathbb{R}^n)$ with

$$Dv = \gamma |u|^{\gamma-2} u \, Du. \qquad |Dv| = \gamma |u|^{\gamma-1} |Du|$$

$$p^* = \frac{np}{n-p}$$

$$1^* = \frac{n}{n-1}$$

We have

$$\left(\int_{\mathbb{R}^n} |v|^{n/(n-1)} dx \right)^{(n-1)/n} \leq C \int_{\mathbb{R}^n} |Dv| \, dx$$

$$\left(\int_{\mathbb{R}^n} |u|^{\overbrace{\gamma n/(n-1)}^{p^*}} dx\right)^{(n-1)/n} \leq C \int_{\mathbb{R}^n} \gamma |u|^{\gamma-1} |Du| \, dx$$

$$\leq C \left(\int_{\mathbb{R}^n} |u|^{\overbrace{(\gamma-1)p/(p-1)}^{p^*}} dx\right)^{(p-1)/p} \left(\int_{\mathbb{R}^n} |Du|^p dx\right)^{1/p}$$

$$\left|\int fg\right| \leq \left(\int |f|^p\right)^{1/p} \left(\int |g|^q\right)^{1/q}$$

$$\frac{1}{p} + \frac{1}{q} = 1$$

$$q = \frac{p}{p-1}$$

Choose γ so that $\dfrac{\gamma n}{n-1} = \dfrac{(\gamma-1)p}{p-1}$:

$$\cancel{\gamma np} - \gamma n = (\gamma - 1)(n-1)p = \cancel{\gamma np} - np - \gamma p + p$$

$$\gamma(n-p) = (n-1)p$$

$$\gamma = \frac{(n-1)p}{n-p}$$

$$\frac{\gamma n}{n-1} = \frac{np}{n-p} = p^*$$

$$\frac{n-1}{n} - \frac{p-1}{p} = \frac{1}{p} - \frac{1}{n} = \frac{n-p}{np} = \frac{1}{p^*}$$

$$\left(\int_{\mathbb{R}^n} |u|^{p^*} dx\right)^{1/p^*} \leq C \left(\int_{\mathbb{R}^n} |Du|^p dx\right)^{1/p} \#$$

<u>Thm:</u> Let $U \subset \mathbb{R}^n$ be a bdd open set with $\partial U \in C^1$ and let $1 \leq p < n$. Then

$$\|u\|_{L^{p^*}(U)} \leq C \|u\|_{W^{1,p}(U)} \quad \forall u \in W^{1,p}(U)$$

for some constant $C > 0$.

<u>Pf:</u> Since $\partial U \in C^1$, u can be extended to a function $\bar{u} \in W^{1,p}(\mathbb{R}^n)$ st $\bar{u} = u$ a.e. in U, $\bar{u}$ has compact support, and

$$\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(U)}$$

for some constant $C > 0$ (independent of u). Since $\bar{u}$ has compact support, $\exists$ a seq (u_j) in $C_c^\infty(\mathbb{R}^n)$ st $u_j \to \bar{u}$ in $W^{1,p}(\mathbb{R}^n)$.

By the last thm, $u_j \in L^{p^*}(\mathbb{R}^n)$ and

$$(*) \quad \underbrace{\|u_j\|_{L^{p^*}(\mathbb{R}^n)}}_{\to \|\tilde{u}\|_{L^{p^*}(\mathbb{R}^n)}} \leq C \|Du_j\|_{L^p(\mathbb{R}^n)} \leq C \underbrace{\|u_j\|_{W^{1,p}(\mathbb{R}^n)}}_{\to \|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)}} \quad \forall j$$

for some constant $C > 0$. We also have

$$\|u_j - u_k\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|u_j - u_k\|_{W^{1,p}(\mathbb{R}^n)} \to 0 \text{ as } j,k \to \infty,$$

so (u_j) is a Cauchy seq in $L^{p^*}(\mathbb{R}^n)$ and hence converges to some $\tilde{u} \in L^{p^*}(\mathbb{R}^n)$.

Since $u_j \to \tilde{u}$ in $L^{p^*}(\mathbb{R}^n)$, a <u>renamed</u> subseq of u_j converges to $\tilde{u}$ a.e. Then $u_j \to \bar{u}$ in $W^{1,p}(\mathbb{R}^n)$ and hence also in $L^p(\mathbb{R}^n)$. So u_j has a further subseq (u_j) that converges to $\bar{u}$ a.e. Since this subseq also converges to $\tilde{u}$ a.e., $\tilde{u} = \bar{u}$ a.e. So passing to the limit in $\circledast$ gives

$$\underbrace{\|\bar{u}\|_{L^{p^*}(\mathbb{R}^n)}}_{} \leq C\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \quad \#$$

$$\|u\|_{L^{p^*}(U)} \overset{?}{\leq} C\|u\|_{W^{1,p}(U)}$$

$$\bar{u} = u \text{ a.e. in } U$$

$$\left(\int_{\mathbb{R}^n} |\bar{u}|^{p^*}\right)^{1/p^*} \geq \left(\int_U |\bar{u}|^{p^*}\right)^{1/p^*} = \left(\int_U |u|^{p^*}\right)^{1/p^*}$$

$$\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(U)}$$

<u>Cor:</u> Let $U \subset \mathbb{R}^n$ be a bdd open set with $\partial U \in C^1$ and let $1 \le p < n$. Then for any $q \in [1, p^*]$, $\exists$ a constant $C > 0$ st

$$\|u\|_{L^q(U)} \le C \|u\|_{W^{1,p}(U)} \quad \forall u \in W^{1,p}(U).$$

<u>Pf:</u> Since U is bdd and $q \le p^*$,

$$\|u\|_{L^q(U)} \le C \|u\|_{L^{p^*}(U)} \le C \|u\|_{W^{1,p}(U)}. \quad \#$$

Pf: $u \in L^q(U)$

$$\int_U |u|^p \, dx = \int_U |u|^p \cdot 1 \, dx$$

$$\leq \left(\int_U (|u|^p)^{q/p} \, dx \right)^{p/q} \left(\int_U 1^{(q/p)^*} \, dx \right)^{1-p/q}$$

$$= \left(\int_U |u|^q \, dx \right)^{p/q} \mu(U)^{1-p/q}$$

$U \subset \mathbb{R}^n$ bdd

$1 \leq p < q < \infty$

$L^p(U) \supset L^q(U)$

$\|u\|_{L^p(U)} \leq \|u\|_{L^q(U)} \, \mu(U)^{1/p - 1/q}$

Thm: Let $U \subset \mathbb{R}^n$ be a bdd open set and let $1 \leq p < n$.

Then for any $q \in [1, p^*]$, $\exists$ a constant $C > 0$ st

$$\underbrace{\|u\|_{L^q(U)} \leq C \|Du\|_{L^p(U)}}_{\text{(Poincare inequality)}} \quad \forall u \in W_0^{1,p}(U).$$

<u>Pf:</u> Since $W_0^{1,p}(U)$ is the closure of $C_c^\infty(U)$, for any $u \in W_0^{1,p}(U)$, $\exists$ a seq (u_j) in $C_c^\infty(U)$ st $u_j \to u$ in $W^{1,p}(U)$.

Extend each u_j to be zero outside U. Then

$$\|u_j\|_{L^{p^*}(U)} \leq C \|Du_j\|_{L^p(U)}$$

by GNS. Pass to the limit to get

$$\|u\|_{L^{p^*}(U)} \leq C \|Du\|_{L^p(U)}. \quad \#$$